

FORMULA MONTHLY

NOVEMBER 2025



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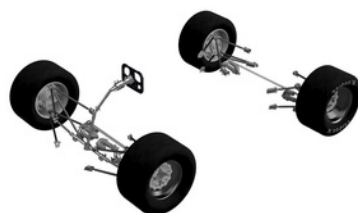
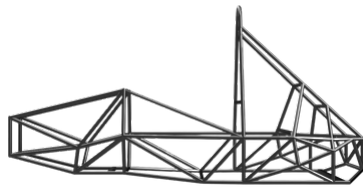


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Featured Vehicle: The All-New Fauget

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PROGRESS**



Structures

completed key validation and finishing tasks, including fabrication of the torsion jig and experimental torsion testing of the chassis, ACP analysis of an initial monocoque iteration, and development of advanced monocoque versions. In parallel, critical fabrication milestones were achieved such as rewelding transmission brackets, welding the AIP, painting the chassis and pushjack, and welding all remaining brackets, bringing the structure close to final completion

Aerodynamics and composites

The Aerodynamics & Composites team successfully completed key design, analysis, and manufacturing milestones, including CFD simulations and validation of multiple aerodynamic concepts, completion of rear wing (RW) mainplane and side element layouts, corrected endplate fabrication and integration, and full RW assembly with verified alignment. Structural and aerodynamic performance targets were met, with downforce values achieved and verified through simulation and checks. Several composite manufacturing processes were finalized, including mold preparation, surface finishing, and curing, while design iterations were closed out following validation. Overall, the completed tasks reflect steady progress from concept evaluation through manufacturing execution and final assembly readiness.

Vehicle dynamics

The team completed key milestones including tire model simulation and comparison, fixing hub offsets, redesign of the front bell crank, and development of the first iteration of the rear upright. Brake rotors were machined, full-vehicle simulations were initiated using Simscape Multibody, and preparations began for dynamic testing and final integration of the car

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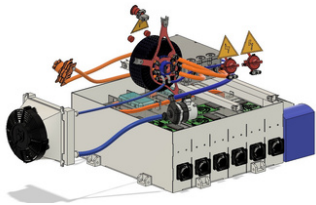
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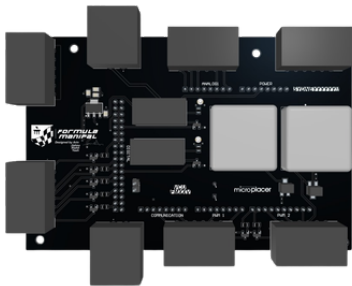
Transmission

The team hit key fabrication milestones, completing final welding and installation of the scatter shield panels. We've selected ISO VG 68 for gearbox lubrication to prioritize thermal stability. Simultaneously, transmission design is nearing finalization, focusing on Emrax 208 integration and refined gear geometry.



E-Powertrain

Our team completed key finishing and consolidation tasks, including design of a backup aluminium ECS compartment, creation of the EPT BOM, replacement of failed 3D-printed segment parts, and redesign of the rear E-box. Safety and integration work progressed with updated busbar insulation, application of ESO and HV stickers, and preparation for final HV wiring and ECS integration



Electronics and controls

The team completed key debugging and validation tasks, including handcart wiring, pressure sensor configuration, resolution of motor shuddering, and testing of the new AMS and latch PCBs. In parallel, telemetry and CAN-based data systems were refined using Teensy-ESP32 communication, and wiring harness insulation and finishing were carried out, moving the electronics system closer to full reliability and integration



Driverless

The team completed key hardware and software milestones, including installation of pedal rails, camera mounts, and ECS components, along with validation of Jetson interfaces and sensors. In parallel, ROS2-based perception and control pipelines were refined through improved LiDAR, camera, and VectorNav integration, advancing the driverless system toward stable vehicle-level operation



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